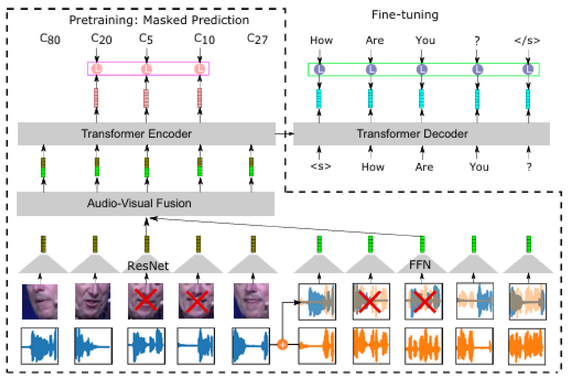
Context-Aware Lip Reading

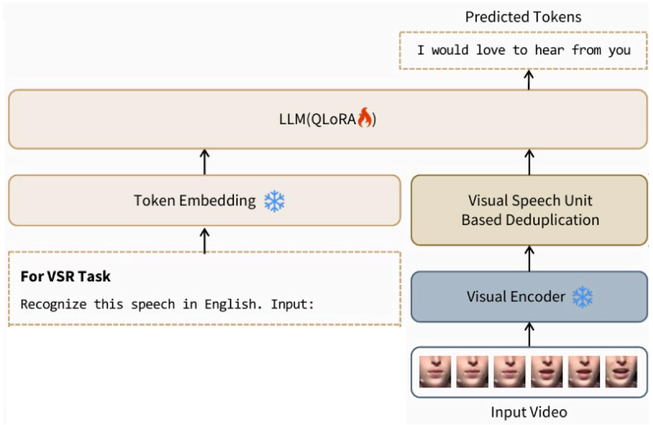
Yuran Zou, Tianyu Cao

Previous Works

Original AV-HuBERT[1]



VSP-LLM (AV-HuBERT + LLaMA)[2]



Core Idea

Context of the speech can be very useful when guessing the content.

Context can include topic, metadata, or general visual features from the "no-crop" video.

Context can be injected into the LLM-decoder in the modality of extra text or video feature tokens.

Implementation

Our code is largely adapted from the VSP-LLM official repository. Especially, we used the same deduplication trick proposed in VSP-LLM to reduce the number of AV-HuBERT tokens.

The AV-HuBERT features and the video features (when existing) are projected into the LM's 4096 dimension embedding space by a simple linear layer. They are then simply concatenated to the text tokens.

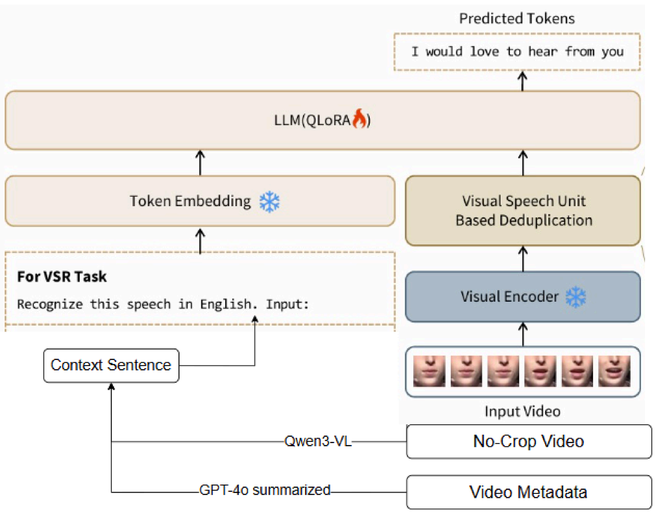
We trained the model for 10000 steps on 2 A6000 GPUs (resp. 4 on the LRS2 dataset) with batch size 1 and 8 gradient accumulation steps.

The AV-HuBERT encoder is frozen during the first 3200 updates.

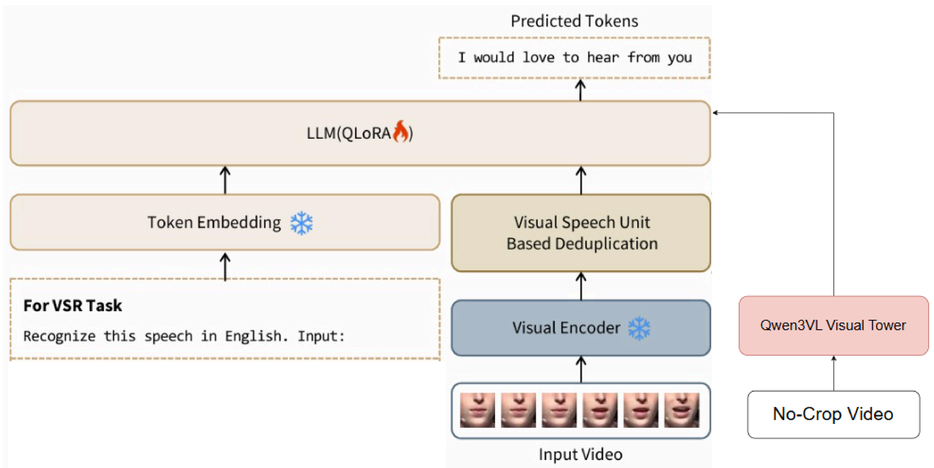
The learning rate is 5e-4, and is warmed up for the first 3000 steps, and decays after 8000 steps.

We save the model each 1000 steps, and pick the checkpoint that performs best on the valid set.

Our Methods



Method 1:
Generate Context Sentence, and insert it into prompt.



Method 2:
Directly pass Qwen3VL-extracted feature from the "no-crop" video.

Github

On the way...

[1] Shi *et al.*, Robust Self-Supervised Audio-Visual Speech Recognition, in INTERSPEECH 2022. [2] Yeo et al., Where Visual Speech Meets Language: VSP-LLM Framework for Efficient and Context-Aware Visual Speech Processing, in Findings of EMNLP 2024. [3] Bai et al., Qwen3-VL Technical Report, arXiv 2025. [4] Prajwal et al., Scaling Multilingual Visual Speech Recognition, in ICASSP 2025. [5] Chung et al., Lip Reading Sentences in the Wild, in CVPR 2017.

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Datasets

We use a subset (4165 youtube IDs) of the English part of MultiVSR[4] as the main dataset.

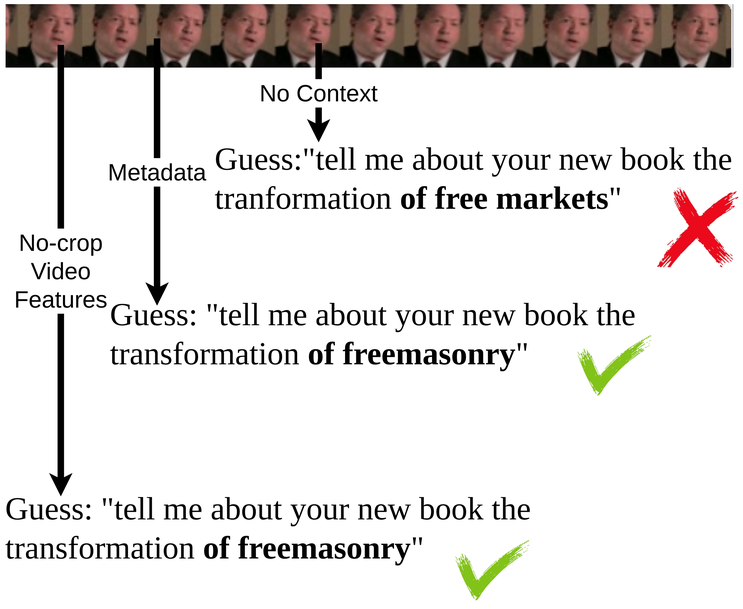
The dataset provides youtube IDs, face tracks and transcription.

We take 5% of videos as validation set and 5% as test set. We filter to keep only 1s~20s clips for train and valid set, and only 1s~6s clips for test set.

Early proof-of-concept uses the LRS2[5] dataset.

Qualitative Results

Ground Truth:
"tell me about your new book the transformation **of freemasonry**"



Quantitative Results

Metric: Word Error Rate (WER) ↓

LRS2

Context Setting	Result
No context (baseline)	24.51
Topic Sentence from GT	**22.40**
5 Keywords from GT	**22.38**

MultiVSR

Context Setting	Result
No context (baseline)	58.24
Youtube Metadata	**55.93**
Topic Sentence from video	**56.96**
Topic Sentence from video & Youtube Metadata	**56.94**
Qwen3VL visual features	**56.28**
Qwen3VL visual features (Qwen3VL frozen)	**56.27**

Note: The LRS2 does not have metadata or no-crop video. As early proof-of-concept, context info is generated from Ground Truth.

Observations:

All kinds of contextual information help the model guess better.

Qwen-generated Topic Sentence does not always have useful clues for generation.

Passing all video features (video encoder frozen or not) is cheaper and slightly better than passing synthesized Topic Sentence.

Notes

We tried changing from LLaMA-7B to Qwen3VL-8B[3] to inject video in a "pretrained" way. However, this prohibits learning (early plateau).

Same phenomemon for Qwen3VL's LM without video input, proving that the problemis in Qwen3's LM, not the visual encoder.

This is why we monkey-patched Qwen3VL'svisual branch with the original LLaMA LM.

Conclusions

Injecting contextual information can consistently improve the LLM-based lip reading model's accuracy.

Contributions

Yuran Zou: adapts the VSP-LLM code, runs the experiments.
Tianyu Cao: Literature research, dataset collection and processing, context data generation.
We thank Wang Xi and Vicky Kalogeiton for their guidance during the project. Generative AIs contributed during coding.

Next Steps

Analysis on Qwen3's language model?
Why it fails, how do different models learn new modalities on small datasets?

Possible biases?
Are the context features encoding speech patterns related to gender or skin tone?

Better architecture for long sentences?
How to mitigate against autoregressive LM's error propagation for long sentences?

[1] Shi *et al.*, Robust Self-Supervised Audio-Visual Speech Recognition, in INTERSPEECH 2022. [2] Yeo et al., Where Visual Speech Meets Language: VSP-LLM Framework for Efficient and Context-Aware Visual Speech Processing, in Findings of EMNLP 2024. [3] Bai et al., Qwen3-VL Technical Report, arXiv 2025. [4] Prajwal et al., Scaling Multilingual Visual Speech Recognition, in ICASSP 2025. [5] Chung et al., Lip Reading Sentences in the Wild, in CVPR 2017.